

TOPICAL REVIEW

Wideband ultrafast fiber laser sources for OCT and metrology

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Topical Review

Wideband ultrafast fiber laser sources for OCT and metrology

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**Abstract**

Fiber lasers, which use optical fibers as ideal waveguides, have been attracting a great deal of attention as stable, practical, and maintenance-free lasers. Using a combination of an ultrashort-pulse fiber laser and a nonlinear fiber, we can realize wideband highly functional ultrafast fiber laser sources. The generation of ultrashort pulses with wide wavelength tunability and supercontinua based on fiber lasers and nonlinear fibers has been demonstrated. These techniques are useful for laser applications, especially for imaging and metrology. In this topical review, the fundamentals of and recent progress in wideband ultrafast fiber laser sources and their applications are reviewed mainly based on the author's work. First, a new pulse source based on a passively-mode-locked ultrashort-pulse fiber laser using carbon nanotubes is explained. Next, the development of wideband ultrafast fiber laser sources and their applications in ultrahigh-resolution optical coherence tomography, optical frequency combs, and nonlinear microscopy are reviewed.

Keywords: ultrashort-pulse fiber lasers, carbon nanotube, ultrafast nonlinear fiber optics, soliton, supercontinuum, optical coherence tomography

(Some figures may appear in colour only in the online journal)

1. Introduction

Fiber lasers, in which the laser cavity consists of an optical fiber, have been attracting a great deal of attention recently as practical lasers. Since an optical fiber is a low-loss, ideal waveguide, fiber lasers have many advantages compared with other lasers, such as high energy efficiency, high point stability, high beam quality, and high gain. A fiber amplifier consists of a few meters of fiber which has a large surface area, giving high heat-radiation efficiency. The fiber is easily bent with a small curvature, and therefore, the fiber laser can be made very compact.

Fiber lasers are also well-matched with highly nonlinear fibers. In optical fibers, although the nonlinear refractive index itself is not so large, since we can confine the light beam inside the small core area along the long propagation length, we can induce a large nonlinear effect. By introducing the output from the fiber laser into specialty fibers, we can construct a wideband light source through such nonlinear effects.

Wideband fiber laser sources have been investigated by several groups (Nishizawa 2009a, Fermann and Hartl 2013, Xu and Wise 2013). These light sources are useful and practical, and their applications have also been investigated recently.

In this topical review paper, fiber-based wideband ultrafast fiber laser sources and their applications are reviewed. First, passively-mode-locked ultrashort-pulse fiber lasers, especially using carbon nanotubes, are reviewed. Next, a wideband wavelength-tunable ultrashort-pulse source is reviewed. Ultrashort-pulse generation with a soliton self-frequency shift induced by intra-pulse stimulated Raman scattering is explained. Using the soliton self-frequency shift, we can demonstrate a rapid and widely wavelength-tunable ultrashort-pulse source. The investigation of ultrashort-pulse generation at shorter wavelengths is also reviewed briefly. Then, a widely broadened supercontinuum (SC) is reviewed, mainly in terms of fiber-laser-based systems. These wideband fiber light sources have been used for many applications. In particular, the application of supercontinua to ultrahigh-

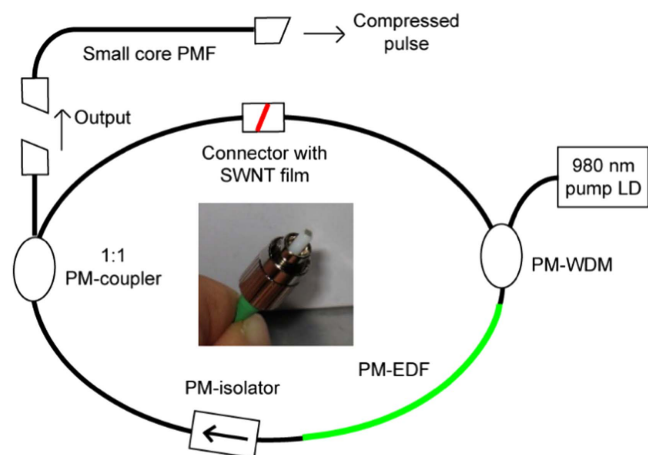


Figure 1. All-polarization-maintaining Er-doped fiber laser with SWNT film. WDM, wavelength division multiplexed coupler; EDF, Er-doped fiber.

resolution (UHR) optical coherence tomography (OCT) is explained. Finally, other applications of wideband fiber laser sources are reviewed.

2. Ultrashort-pulse fiber laser sources

Passive mode-locking is used in fiber lasers to generate ultrashort pulses. Intensity-dependent elements are used in the cavity to achieve passive mode-locking. At the early stages of the development of fiber lasers in the 1990s, passive mode-locking methods using the nonlinear phase shift induced by the optical Kerr effect were developed. A figure-eight cavity with a nonlinear amplifying mirror and nonlinear polarization rotation have been used to achieve passive mode-locking (Duling 1991, Matsas *et al* 1992, Agrawal 2008). Intensity-dependent loss was induced through the interference of nonlinear phase-shift components, making the response speed almost instantaneous and also realizing a wide operating bandwidth. In principle, this mode-locking method does not depend on the wavelength, and it can work in any wavelength region.

Passive mode-locking is easily achieved using a saturable absorption device. A semiconductor saturable absorption mirror (SESAM) is one example of a suitable saturable absorber. Using a SESAM, the self-start operation of passive mode-locking was achieved with low pulse energy. SESAMs have been used in many commercial fiber laser products (Loh 1993, Fermann *et al* 2003).

Recently, nano-carbon materials have been attracting a great deal of attention as new saturable absorbers. The saturable absorption property of single wall carbon nanotubes (SWNTs) was discovered in 2002 (Chen *et al* 2002). Subsequently, the first demonstration of an ultrashort-pulse fiber laser using SWNTs was reported in 2003 (Set *et al* 2003). SWNTs show a wide operating bandwidth and fast response time, typically ~ 1 ps. Transmission-type devices can be fabricated using SWNTs, and it is easy to set them inside the

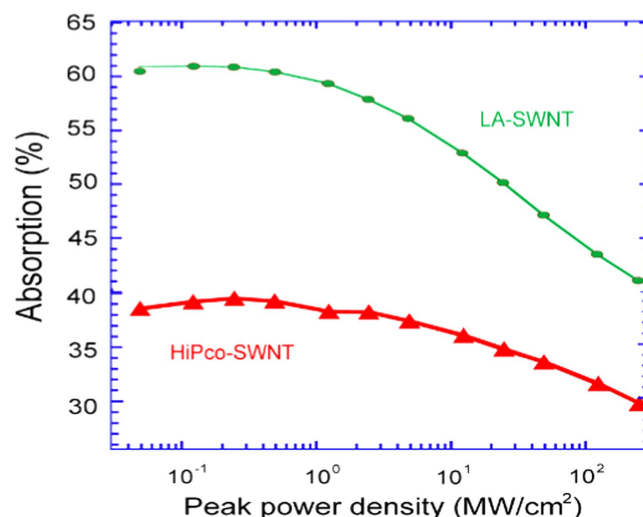


Figure 2. Observed saturable absorption properties of SWNT polyimide film.

fiber laser cavity. Several kinds of SWNT devices have been fabricated, including a direct deposition type, a film type, and a tapered fiber type (Yamashita *et al* 2004, Sakakibara *et al* 2005, Kieu and Mansuripur 2007).

We have been investigating Er-doped ultrashort-pulse fiber lasers using SWNTs. As the mode-locker, we have been using a polyimide film device in which SWNTs are dispersed. SWNTs synthesized by the high-pressure CO (HiPco) method and the laser ablation (LA) method were used. SWNTs with a diameter of 1.2 nm show saturable absorption in the $1.55 \mu\text{m}$ wavelength region. Using the LA method, SWNTs with a diameter of 1.2 nm can be selectively synthesized, and large modulation depth can be obtained. In this method, the SWNTs were purified and then dispersed in polyimide liquid. A film-type device was fabricated after the evaporation. This film-type SWNT device works as a transmission-type saturable absorber, and it is useful for demonstrating all-polarization-maintaining fiber lasers as shown in figure 1 as shown in figure 1 (Nishizawa *et al* 2008a, Senoo *et al* 2009).

Figure 2 shows the observed saturable absorption properties of the developed SWNT polyimide films. As the peak power density of the irradiation pulse was increased, the magnitude of absorption decreased owing to the saturation of absorption. Thanks to the high selectivity of the LA method, a large modulation depth of $\sim 20\%$ was observed for the SWNTs synthesized by this method. For the SWNTs synthesized by the HiPco method, the observed modulation depth was $\sim 10\%$.

Figure 3 shows the variation of the output power and operating mode as a function of pump power. As the pump power was increased past the threshold power, the output power increased almost linearly. The operating mode shifted from continuous wave (CW) oscillation to self-Q switching, and then single-pulse mode-locking. Self-start single-pulse mode-locking was achieved only by increasing the pump power. The maximum power of single-pulse oscillation was

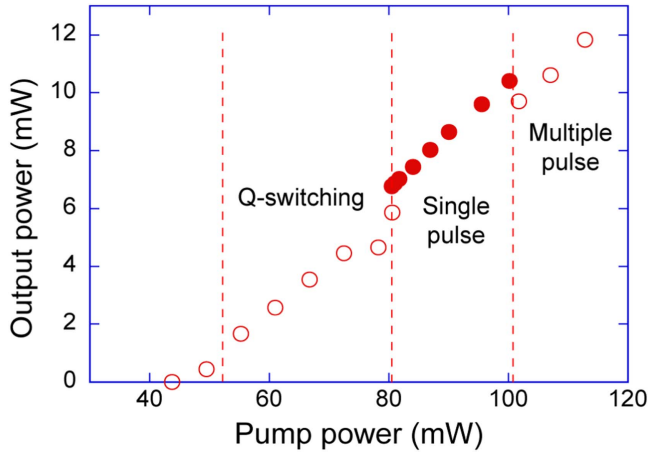


Figure 3. Output power and operating mode as a function of pump power. Adapted from Nishizawa 2008a. Copyright OSA 2008.

limited by the single-pulse oscillation limit. Above this limit, multiple-pulse oscillation occurred.

Figure 4 shows the numerical results of passive mode-locking in an SWNT fiber laser (Nishizawa *et al* 2015). The extended nonlinear Schrodinger equation was used for the analysis. The amplified spontaneous emission (ASE) noise acted as the seed beam, and it circulated around the fiber laser cavity. The ASE noise underwent amplification in the EDF and intensity-dependent saturable absorption at the SWNT polyimide film. As a result, an intense temporal component gradually survived, and it was gradually amplified while circulating. Then, pulse compression occurred through soliton compression, increasing the peak power and decreasing the temporal width, and then stable pulse oscillation was achieved. In the case of soliton mode-locking, in which the magnitude of net dispersion is negative, a sech^2 -shaped ultrashort pulse was generated stably at the fiber laser output. As shown in figures 4(b) and (c), an almost chirp-free clean sech^2 -shaped pulse was generated.

By changing the dispersion property of the fiber laser cavity, we can obtain different mode-locking regimes. Figure 5 shows the configuration of a dispersion-managed EDF laser with an SWNT film (Nishizawa *et al* 2011a). Normal-dispersion fiber (NDF) was used to control the magnitude of net dispersion. A variable coupler was used as the output coupler, and an asymmetric branching ratio was used to suppress the irradiation power on the SWNT film and to achieve a high output power.

Figure 6 shows the variation of the laser output as a function of total dispersion D_T . When $D_T < 0$, soliton mode-locking operation was obtained, and a stable sech^2 -shaped ultrashort pulse was generated. In this regime, however, the energy of a single pulse was limited by the single-pulse oscillation limit. Figure 6(b) shows the observed pulse spectrum and spectral phase of the output pulse when $D_T = -0.032 \text{ ps}^2$. An almost chirp-free clean sech^2 -like pulse was obtained.

As D_T was increased, the oscillation mode shifted from soliton mode-locking to a stretched pulse, and then dissipative

soliton mode-locking operation. The average power and temporal width increased as D_T was increased. In particular, a high pulse energy could be obtained in the dissipative soliton mode-locking regime. A pulse spectrum with steep edges at both sides and a linear chirping property, as shown in figure 6(c), was observed in the dissipative soliton mode-locking regime. Using this regime, we achieved the highest reported output power of 285 mW for a single-pulse mode-locked nano-carbon fiber laser (Nozaki *et al* 2012).

In principle, the stretched-pulse mode-locking operation is achieved when $D_T \sim 0$. In our work, however, stable mode-locking was not obtained when $D_T \sim 0$. It is considered that this is due to the limited modulation depth of the SWNT film. Recently, stretched-pulse mode-locking operation has been reported for SWNT fiber lasers using modified cavity configurations (Kwon *et al* 2015, Wang *et al* 2015).

Graphene is another attractive nano-carbon material which shows saturable absorption properties. Thanks to the unique bandgap structure, it works in a very wide wavelength region. Several groups have been investigating ultrashort-pulse fiber lasers using graphene (Nonaccorso *et al* 2010, Yamashita 2012, Martinez and Sun 2013). Recently, other novel materials, such as hexagonal boron nitride (hBN), topological insulator, and transition metal dichalcogenides (TMDCs), have also been investigated as new saturable absorbers by several groups (Neto and Novoselov 2011, Sobon *et al* 2012, Xu *et al* 2013, Mao *et al* 2015).

3. Wideband ultrafast fiber sources

Fiber lasers are stable, practical, and useful laser sources. However, the oscillation wavelength is limited within the gain bandwidth of the fiber amplifier. For applications that have wavelength dependence, wavelength tuning or spectral broadening techniques are important.

Using the fiber nonlinear effects, we can demonstrate wideband ultrashort-pulse fiber laser sources. When an intense ultrashort pulse propagates along an anomalous-dispersion fiber, pulse compression occurs by the soliton effect, and the pulse spectrum is broadened. Then, the longer wavelength side is enhanced by stimulated Raman scattering inside the pulse spectrum, and the longer wavelength side breaks off from the original pulse component. This allows a red-shifted ultrashort soliton pulse to be generated through stimulated Raman scattering and the soliton effect. During propagation, the center wavelength of the soliton pulse is continuously shifted towards the longer wavelength side by intra-pulse Raman scattering (stimulated Raman scattering inside the pulse), while the pulse shape maintains a sech^2 -shape by the soliton effect. The magnitude of the wavelength shift depends on the input pulse power and fiber length, and a wavelength-tunable ultrashort soliton pulse can be generated. This phenomenon is called the 'soliton self-frequency shift', which was discovered experimentally by Mitschke and Mollenauer in 1986 (Mitschke and Mollenauer 1986). In those experiments, picosecond pulses and kilometer long fiber were used. Later, in 1999, Nishizawa and Goto demonstrated

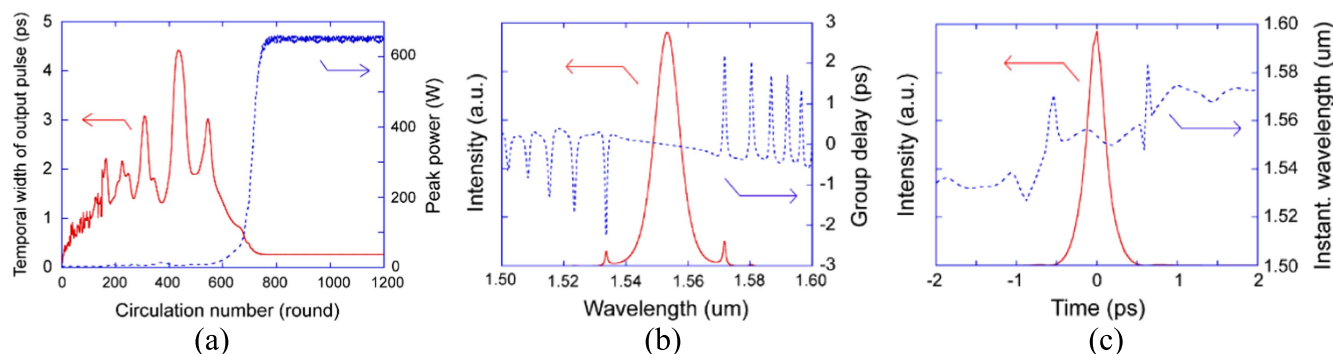


Figure 4. Numerical results of passive mode-locking in a soliton mode-locked fiber laser. (a) Variation of pulse width and peak power in the initial process of passive mode-locking, (b) pulse spectrum and group delay, and (c) temporal shape and instantaneous wavelength of output pulse after mode-locking.

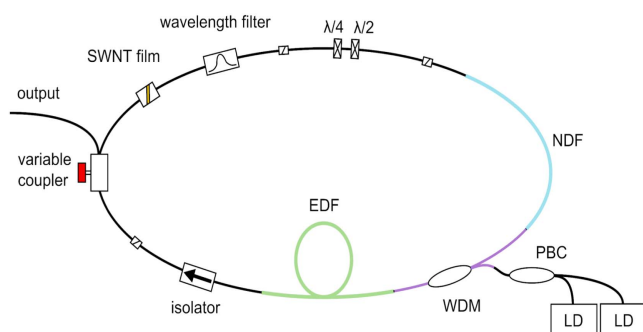


Figure 5. Configuration of a dispersion-managed ultrashort-pulse EDF laser with SWNT film. PBC, polarization beam combiner.

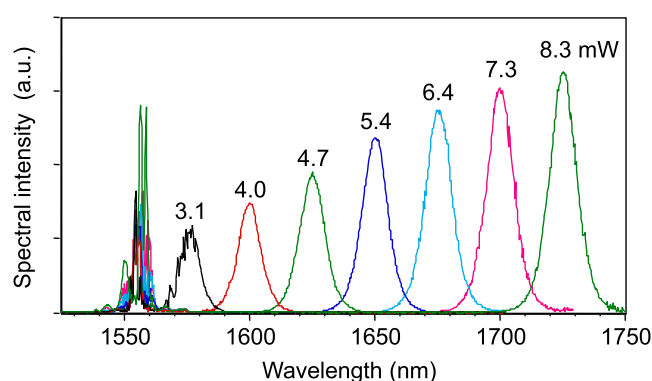


Figure 7. Output spectra of wavelength-tunable soliton pulses.

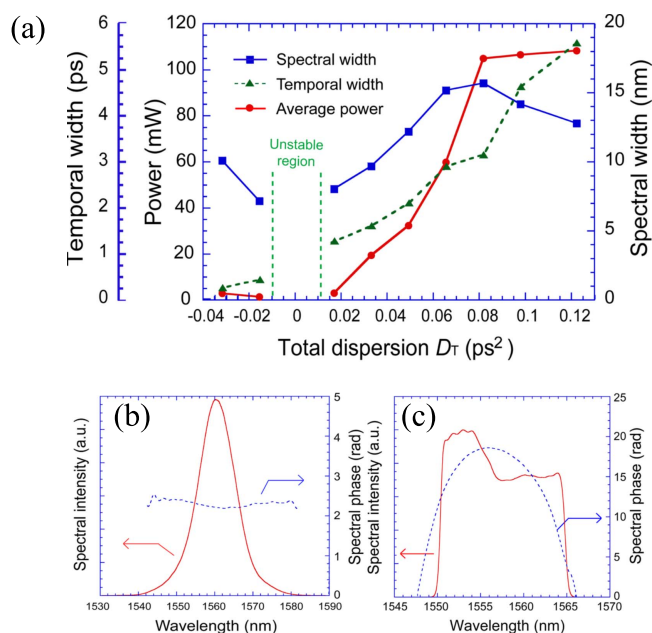


Figure 6. Experimental results of the output characteristics of a dispersion-managed SWNT fiber laser: (a) Output power and operating mode as a function of pump power, and spectral shapes of output pulse for (b) soliton mode-locking regime, and (c) dissipative soliton mode-locking regime. Reproduced with permission from Nishizawa 2011a. Copyright OSA 2011.

wideband wavelength tuning using a combination of an ultrashort-pulse fiber laser and an optical fiber (Nishizawa and Goto 1999a).

Figure 7 shows the optical spectra of wavelength-tunable ultrashort soliton pulses. An Er-doped ultrashort-pulse fiber laser was used as the seed-pulse source, and the output pulses were coupled into polarization-maintaining single-mode fiber with a small core and anomalous-dispersion properties. As the fiber input power was increased, pulse breakup occurred, and a new pulse was generated at the longer wavelength side of the pump pulse. A clean sech^2 -shaped soliton pulse was generated. As the fiber input power was increased, the center wavelength of the soliton pulse was red-shifted continuously, while keeping the ideal pulse shape. Using a high-power Er-doped ultrashort-pulse fiber laser and a small-core polarization-maintaining fiber, a wide wavelength shift from 1.55–2.0 μm was achieved (Nishizawa *et al* 1999b).

Using the birefringence of a polarization-maintaining fiber, we can generate two-color soliton pulses simultaneously (Nishizawa *et al* 1999c). Then, the wideband wavelength-tunable soliton pulse generation has been investigated using photonics crystal fibers and several different ultrashort-pulse sources, such as optical parametric oscillators (Liu *et al* 2001), Ti:Sapphire lasers (Cormack *et al* 2002, Serbryannikov 2005), and Cr:Forsterite lasers (Chan *et al* 2008).

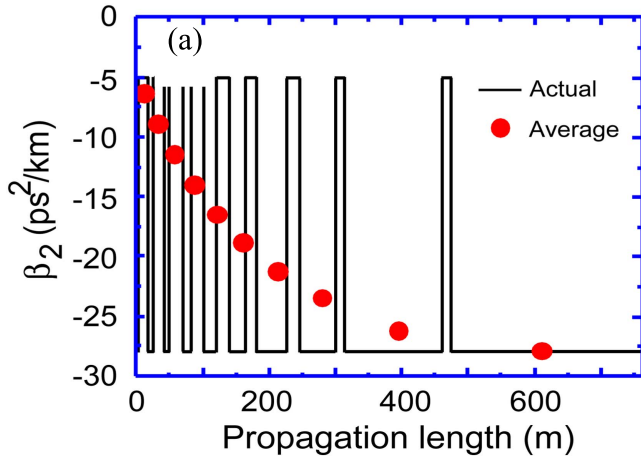


Figure 8. Dispersion map of developed quasi-dispersion-increasing fiber using the comb-profile fiber technique. Reproduced with permission from Nishizawa *et al* 2010. Copyright IOP Publishing 2010.

If we use an Yb-doped ultrashort-pulse fiber laser and a photonic crystal fiber (PCF), we can generate widely wavelength-tunable ultrashort pulses starting from a wavelength of 1.06 μm (Lim *et al* 2004, Takayanagi and Nishizawa 2006a, van Howe *et al* 2007). In our group, the observed longest wavelength for an Yb-doped fiber laser system was as high as 1.9 μm (Sumimura *et al* 2010). The wavelength-tunable soliton pulse generation pumped with Tm-Ho fiber laser was also investigated (Kivisto *et al* 2007). Since the magnitude of the wavelength shift is almost proportional to the fiber input power, we can demonstrate rapid wavelength tuning operation using an intensity modulator. Using a fast intensity modulator with a bandwidth of a few tens of GHz, we can demonstrate ultrafast wavelength tuning. Actually, we can tune the wavelength shot-by-shot. To the best of our knowledge, this is the fastest wideband wavelength-tunable laser system yet reported (Nishizawa and Takahashi 2011b).

Since the generated soliton pulse has to obey the fundamental soliton condition, the following equation is satisfied:

$$N = \frac{\gamma P_0 T_0^2}{|\beta_2|} = 1, \quad (1)$$

where N is the soliton order, γ is the nonlinear coefficient, P_0 is the peak power, T_0 is the pulse duration, and β_2 is the second-order dispersion. From equation (1), we can see that a high-energy soliton pulse can be generated with a large-core fiber, or a small-nonlinearity fiber. Using a PCF with a large mode-field area (LMA), a 104 fs soliton pulse with 10 nJ pulse energy was generated (Nishizawa and Hori 2008b). Using a rod-type PCF, a high-energy soliton pulse with a pulse energy of more than 10 nJ was generated in the 1.7 μm region, and was used for three-photon microscopy deep inside a tissue sample (Horton *et al* 2009).

Since a soliton pulse is an ultrashort pulse with a typical pulse width of 50–200 fs, it inherently has a wide spectral width. If we can compress the spectral width of the generated pulse, we can demonstrate a narrow-linewidth source with wide wavelength tunability.

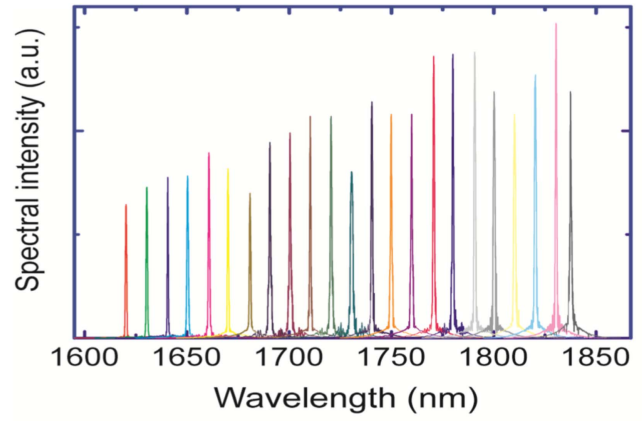


Figure 9. Optical spectra of narrowband and widely wavelength-tunable sources. Reproduced with permission from Nishizawa *et al* 2010. Copyright IOP Publishing 2010.

Recently, we demonstrated high-quality spectral compression using adiabatic soliton spectral compression (Nishizawa *et al* 2010). A quasi-dispersion-increasing fiber was constructed using the comb-profile fiber technique. Figure 8 shows the dispersion map of the developed quasi-dispersion-increasing fiber using the comb-profile fiber technique. Lengths of conventional single-mode fiber and dispersion-shifted fiber were spliced alternately, while changing the ratio of the fiber lengths, and the averaged magnitude of dispersion was set to increase gradually along the fiber. Using the developed comb-profile fiber, the spectral width of a 200 fs soliton pulse was compressed from 12.3–15.1 nm down to 0.54–0.71 nm. A high compression ratio of up to 25.9 was achieved. It is worth noting that the pedestal component, which generally arises during spectral compression, was well suppressed in this scheme thanks to the soliton effect. Thus, ideal spectral compression was demonstrated.

Figure 9 shows the observed spectra of a widely wavelength-tunable narrow-linewidth source. The wavelength-tunable pulse shown in figure 7 was input into a quasi-dispersion-increasing fiber made by the comb-profile fiber technique. A narrow spectrum with small sidebands was generated in a wide wavelength range, and we confirmed the wide operating range of the developed system. Using the electro-optical modulator, we demonstrated a narrow-linewidth source with rapid and wide wavelength tunability (Nishizawa and Takahashi 2011b). Spectral compression using a specially prepared dispersion-increasing fiber has also been reported (Chuang and Huang 2011).

So far, red-shifted soliton pulse generation through the soliton self-frequency shift has been described. In addition to a red-shifted soliton pulse, we can generate a blue-shifted pulse.

In 1995, anti-Stokes pulse generation was reported by Beaud *et al*. In that work, a picosecond pulse was used as the pump pulse source (Beaud *et al* 1987). In 2000, Nishizawa and Goto demonstrated blue-shifted short-pulse generation by coupling an ultrashort pulse into a dispersion-shifted fiber (Nishizawa *et al* 2000). They also analyzed this phenomenon using the cross-correlation frequency-resolved optical gating (X-FROG) technique, and they observed a pulse trapping

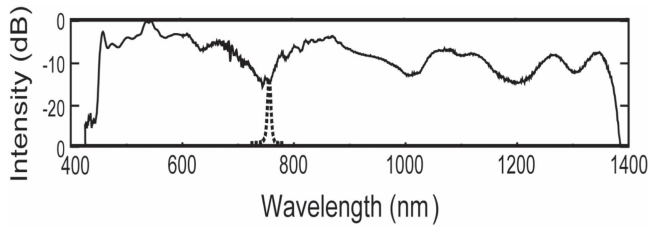


Figure 10. Optical spectra of a 0.4–1.4 μm widely broadened SC.

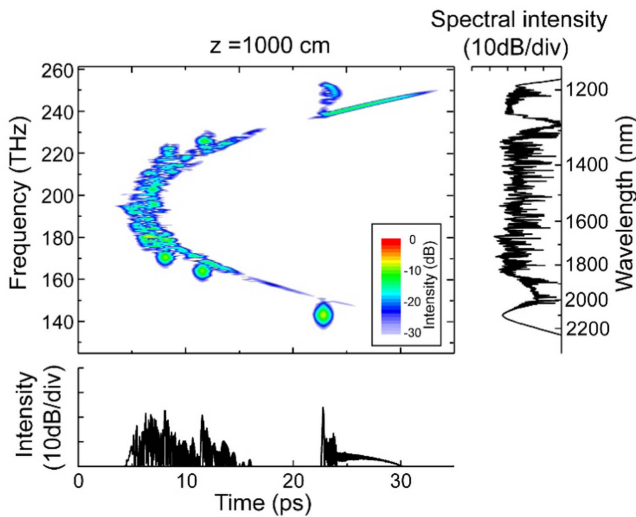


Figure 11. Temporal distribution of spectral components of a widely broadened SC in highly nonlinear dispersion-shifted fiber (numerical results).

phenomenon across the zero-dispersion region (Nishizawa and Goto 2001a, Nishizawa, and Goto 2002). In this phenomenon, the red-shifted soliton pulse traps a pulse at a shorter wavelength through cross-phase modulation. As the wavelength of the soliton pulse becomes more red-shifted, the wavelength of the trapped pulse is blue-shifted along the propagation length. Using this phenomenon, wavelength shifts in the near-infrared and visible regions were demonstrated (Nishizawa and Goto 2002, Nishizawa and Itoh 2009b). Then this trapping phenomenon was also investigated theoretically in terms of a gravity-like trapping force (Gorbach and Skryabin 2007).

In recent work, efficient ultrashort-pulse generation in the visible range using Cherenkov radiation induced in an optical fiber has been demonstrated. A clean ultrashort pulse was generated efficiently using this method (Chang *et al* 2010). Extension of supercontinua towards the visible region using pulse trapping has also been investigated (Stone and Knight 2008, Travers *et al* 2008). Recently, efficient ultraviolet (UV) light generation has been demonstrated with a gas-filled air core photonic bandgap fiber (Travers *et al* 2011).

4. SC generation

A strong nonlinear effect is induced by using a highly nonlinear fiber and an ultrashort pulse at a wavelength around the

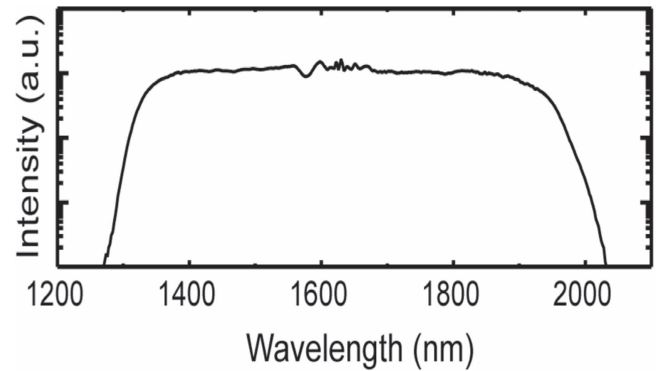


Figure 12. Optical spectrum of a high-quality SC.

zero-dispersion wavelength of the fiber, and this can be used to generate a widely broadened SC. It used to be necessary to use sub- μJ pulse energy to generate an SC in conventional fibers (Alfano and Shapiro 1970, Alfano 2006). However, since the invention of highly nonlinear fibers and PCFs, we can now generate a widely broadened SC with a nJ pulse (Ranka *et al* 2000, Birks *et al* 2000, Nishizawa and Goto 2001b).

Figure 10 shows the observed spectra of a widely broadened SC. The amplified ultrashort pulse from an Er-doped ultrashort-pulse fiber laser was focused into periodically poled LiNbO₃ (PPLN), and a second harmonic generation (SHG) pulse was generated. The generated SHG pulse was coupled into a highly nonlinear PCF whose zero-dispersion wavelength was close to the wavelength of the SHG pulse. As shown in figure 10, a 0.4–1.4 μm widely broadened SC was successfully generated (Nishizawa *et al* 2009c).

Widely broadened SCs are attractive as extremely wideband, bright light sources. Actually, although they are useful for general applications, it has been reported that such wideband SCs have lots of fine structure in the spectra, inherently large noise, and incoherent properties (Gaeta 2002, Gu *et al* 2002). Figure 11 shows the numerically obtained temporal distribution of spectral components of a general widely broadened SC. We can see the fine structures on the spectra. These properties cause serious issues in the applications that need high-quality light sources.

Several groups have investigated SC generation processes using ultrashort pulses and highly nonlinear fibers (Husakou and Herrmann 2002, Orgigosa-Blanch *et al* 2002, Washburn *et al* 2002, Genty *et al* 2002, Coen *et al* 2002, Hori *et al* 2004a, Dudley *et al* 2006, Knight and Skryabin 2007). We also tackled these issues, and we succeeded in developing a method of generating a coherent, low-noise SC (Hori *et al* 2004b)).

Figure 12 shows the optical spectrum of a coherent, low-noise, almost-flat, high-quality SC (Takayanagi *et al* 2006b). The key points are the ideal shape of the pump pulse and the dispersion property of the highly nonlinear fibers. We generated an ideal sech^2 -shaped soliton pulse from an amplified ultrashort pulse through intra-pulse Raman scattering, and

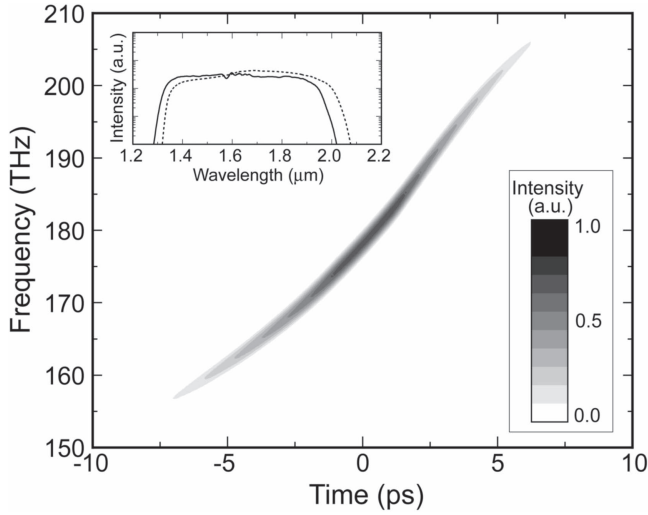


Figure 13. Numerically calculated temporal distribution of the spectral component of the generated high-quality SC. Inset, optical spectra of the numerically simulated SC (broken line) and experimentally observed one (solid line). Reproduced with permission from Nishizawa [2007](#). Copyright OSA 2007.

then introduced it into a normal-dispersion highly nonlinear fiber.

When the ultrashort pulse was coupled into the highly nonlinear fiber around the zero-dispersion region, the spectrum was broadened by self-phase modulation at the initial stage, and then additional broadening occurred through stimulated Raman scattering and four-wave mixing. Temporal overlapping occurred between the broadened spectral components, causing degradation of the spectra (Hori *et al* [2004a](#)).

When the ultrashort pulse was coupled into highly nonlinear fiber with normal-dispersion properties, the spectrum was broadened through only self-phase modulation, and it was temporally broadened monotonically without temporal overlapping. In this case, a smoothly broadened, coherent, high-quality SC was generated.

Figure 13 shows the numerically calculated spectrogram (temporal distribution of spectral components) of a generated SC (Nishizawa and Takayanagi [2007](#)). The inset shows the comparison of optical spectra of a numerically simulated SC and experimentally observed one. Almost the same, smooth SC spectrum was generated from one ultrashort pulse for every shot. For the spectrogram, a monotonous, smooth, and almost linearly broadened spectrogram was observed. We can see that there is no temporal overlapping between the spectral components. Actually, the intensity noise and coherence were examined. We confirmed that the noise level was not increased during the SC generation process, and almost the perfect coherence of 0.8–1.0 was confirmed in the SC for the observed region (Nishizawa and Takayanagi [2007](#)).

Using an LMA PCF, we can generate a high-energy ultrashort soliton pulse. This pulse source was also used for highly sensitive nonlinear microscopy. We generated an octave-spanning, almost-flat SC using this high-energy soliton pulse and highly nonlinear NDF (Nishizawa and Takayanagi [2007](#), Nishizawa and Hori [2008b](#)). Recently, we

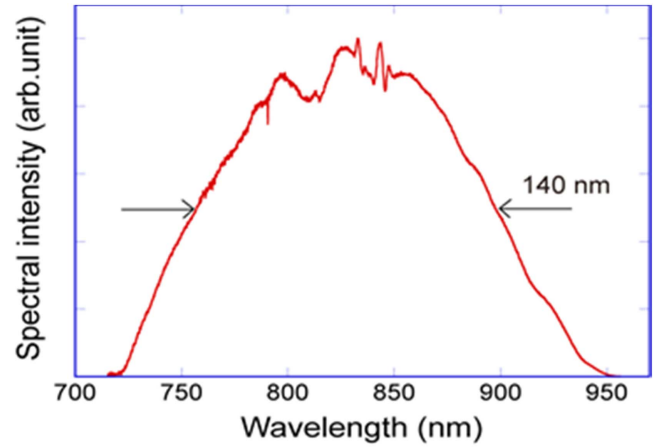


Figure 14. Gaussian-like wideband SC for UHR-OCT.

successfully generated an octave-spanning SC using a 46 fs high-power ultrashort pulse generated with a similariton amplifier (Nozaki *et al* [2014](#)).

Nowadays, many groups have been trying to increase the bandwidth of SC toward the visible, UV, and mid-infrared (MIR) regions. For example, the generation of an SC broadened down to the visible and UV regions was demonstrated using a specially designed PCF (Stone and Knight [2008](#)). Recently, the generation of an ultrawideband SC broadened from UV to 6.28 μm was demonstrated using a fluoride fiber (Qin *et al* [2009](#)).

5. UHR-OCT using wideband fiber lasers

One of the important applications of SC is UHR-OCT (Huang *et al* [1991](#), Bouma and Tearney [2001](#), Drexler and Fujimoto [2015](#)). OCT is a micrometer-scale, non-destructive, cross-sectional imaging technique, based on an incoherent interferometer. The theoretical axial resolution Δz , which corresponds to the coherence length, is written as

$$\Delta z = \frac{2 \ln 2}{\pi} \frac{\lambda^2}{\Delta \lambda} \quad (2)$$

where λ and $\Delta \lambda$ represent the center wavelength and the spectral width of the light source. Since the coherence length is inversely proportional to the bandwidth of the light source, we can demonstrate UHR imaging using a wideband SC. Figure 14 shows the optical spectrum of generated SC for UHR-OCT. There are several approaches for generating a wideband SC. In terms of OCT imaging, in addition to the bandwidth, low noise, high power, and a good spectral shape are required to achieve clear images with high sensitivity.

Figure 15 shows the configuration of an OCT system (Nishiura *et al* [2010](#)). This is the fundamental configuration of an OCT called time-domain OCT, where the interference signal is detected by scanning the reference arm length. The interferometer consists of wideband fiber couplers, a galvanometer scanner, polarization controllers, and lenses. The interference signal is detected at the balanced detector. When a wideband light source is used, dispersion compensation is

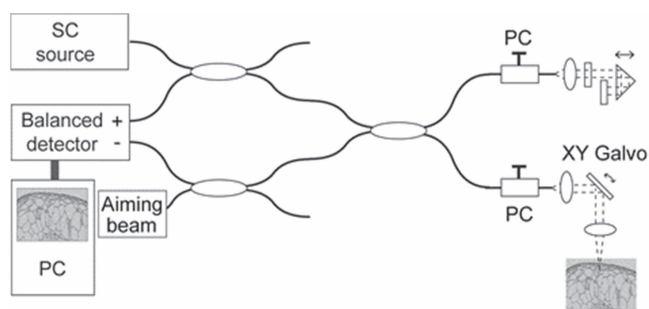


Figure 15. Configuration of a UHR time-domain OCT system. PC, polarization controller.

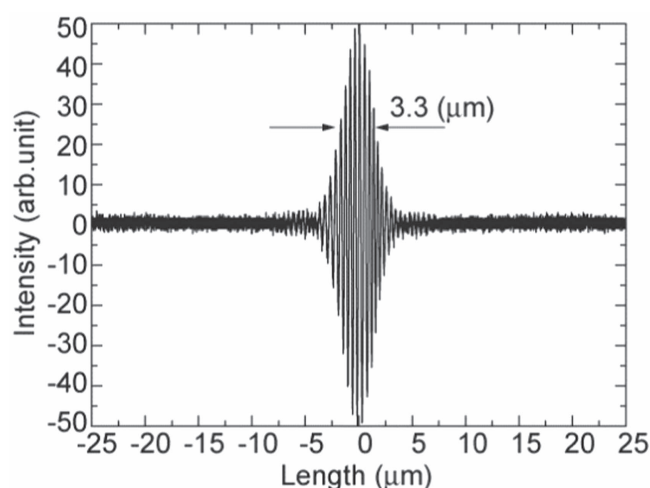


Figure 16. Observed interference signal using an SC.

important for achieving sharp interference signals. The interference signal is detected at the balanced detector, and the obtained electrical signal is acquired with a high-speed analog-to-digital AD board, and a personal computer.

Figure 16 shows the observed interference signal when the SC shown in figure 14 was used (Nishiura *et al* 2010). The observed axial resolution, which is the full width at half maximum (FWHM) of the interference signal, was $3.3 \mu\text{m}$ in air. The corresponding theoretical axial resolution was $3.0 \mu\text{m}$, and it was close to the observed one. Since the signal-to-noise ratio and power were high, a high sensitivity of over 100 dB was achieved.

Figure 17 shows the observed UHR-OCT images of biological samples. Clear images were observed with UHR, and detailed cross-sectional structures of biological samples were observed thanks to the high-power, high-quality SC. Since the absorption of water is low at the wavelength range around 800 nm, this range has been used for ophthalmology, especially observation of the retina.

Imaging speed is one of the important factors, especially for medical applications. Using a monochromator with a camera or a line sensor, we can construct a spectral-domain (SD) OCT system in which interference information is obtained from the spectrum, and fast imaging can be demonstrated (Hattori *et al* 2015).

Figure 18 shows the construction of a UHR-SD-OCT system using an SC. The SC was introduced into a fiber interferometer consisting of a broadband fiber coupler, and spectral interference was observed in the spectrometer. The spectrometer consists of a transmission-type diffraction grating and a fast line-scan camera, enabling fast OCT imaging. The observed interference signal was introduced into a personal computer through a fast AD board, and a cross-sectional image was constructed after signal processing, mainly Fourier transformation. A high axial resolution of $3.1 \mu\text{m}$ in air ($2.3 \mu\text{m}$ in tissue) and a high sensitivity of 104 dB were achieved at an A-scan rate of 70 kHz.

Figures 19(a) and (b) show UHR-OCT images of a zebrafish egg just one day after fertilization. The transformation of the zebrafish embryo was clearly observed. The formation of the eye and head was visible. Small particles inside the yolk were also observed *in vivo*.

Figures 19(c) and (d) show observed cross-sectional images of swimming medaka fish. The average imaging speed was as fast as 15 frames/s. A young medaka was placed in water in a test tube. Since there was nowhere for the medaka fish to escape to, it kept swimming at the same position. As shown in figure 19(c), detailed structures of the brain, such as the torus longitudinalis, optic tectum, and torus semicircularis, were clearly observed. As shown in figure 19(d), the structure of the internal ears was clearly observed.

Generally, the penetration depth of OCT imaging in biological tissue is limited by absorption and scattering. The absorption of hemoglobin at the shorter wavelength side and that of water at the longer wavelength side are well-known dominant factors in biological samples. As for scattering, the magnitude of scattering decreases as the wavelength increases. The magnitudes of absorption and scattering depend on the structure and materials of the tissue being observed, and as a result, the characteristics of OCT imaging depend on the wavelength and sample type.

We have been investigating the wavelength dependence of OCT imaging using OCT systems with SCs at five different wavelengths (Ishida and Nishizawa 2012). Figure 20(a) shows the observed spectra of the generated SCs. A Gaussian-like, low-noise, coherent SC was generated at each wavelength. Figure 20(b) shows the theoretical axial resolution in air as a function of wavelength, and the experimentally observed axial resolution. An ultrahigh axial resolution below $5 \mu\text{m}$ in tissue was observed at each wavelength.

Figure 21 shows the configuration of an SC source operating at a wavelength of $1.7 \mu\text{m}$ (Kawagoe *et al* 2014). An Er-doped ultrashort-pulse fiber laser using a SWNT film was used as the seed-pulse source. It generated ultrashort pulses at a repetition rate of 100 MHz. The generated pulses were amplified in an EDF amplifier, and a wavelength-shifted ultrashort soliton pulse was generated at a polarization-maintaining fiber. Then the residual components were removed at a wavelength filter, and the high-power soliton pulse was coupled into highly nonlinear fiber, generating an SC. Figure 21(b) shows the observed spectrum of the generated SC. An almost Gaussian-like SC with a bandwidth of 242 nm was generated. Since we used a 100 MHz

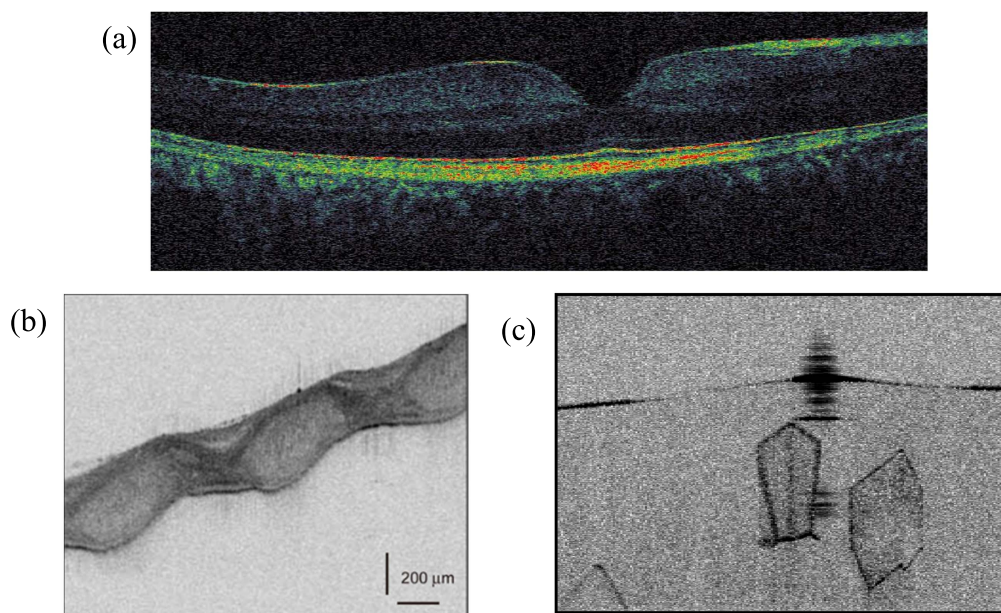


Figure 17. Observed UHR-OCT image of (a) a human retina, (b) a rat trachea, and (c) protein crystals.

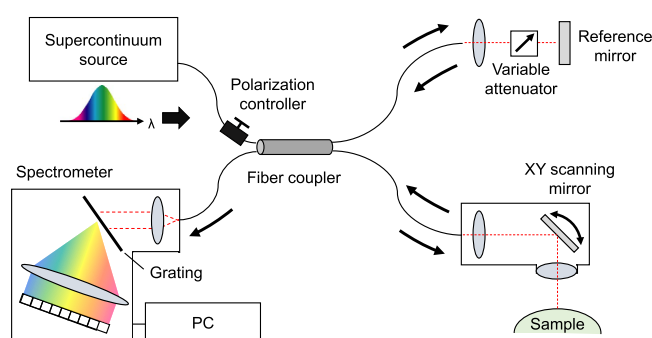


Figure 18. Setup of a spectral-domain UHR-OCT using an SC (Hattori 2015).

high-repetition-rate fiber laser, we obtained a high average power almost twice as large as that of the previous system.

Figure 22 shows 3D UHR-OCT images of a pig trachea and a human baby tooth. Clear images of the samples were observed with UHR. For the OCT images obtained with the short-wavelength source, the precise structures were observed with ultrahigh axial resolution. On the other hand, for the images observed with the longer wavelength source, both high axial resolution and deeper penetration depth were achieved. Generally speaking, a larger penetration depth is achieved as the wavelength is increased for samples with low water absorption. We have been investigating the wavelength dependence of OCT imaging for several different kinds of samples.

Recently, UHR-OCT imaging using commercially available SC sources has been demonstrated by several groups (Zhi *et al* 2011). In addition to cross-sectional structural images, spectroscopic information was examined using an SC with a wide spectral width (Cimalla *et al* 2009, Robles *et al* 2011). Very recently, the advantages of the longer

wavelength region have also been shown for brain imaging (Chong *et al* 2015).

6. Applications of wideband ultrashort-pulse fiber laser sources

A wideband ultrashort-pulse fiber laser is useful as a wideband light source. Since it is based on a passively-mode-locked ultrashort-pulse fiber laser, if the coherence properties are preserved, it can work as a wideband optical frequency comb source. Using a fiber laser system, we can construct a stable and practical all-fiber optical frequency comb system.

In 1999 and 2000, the application of the optical frequency comb generated from passively-mode-locked laser to the optical frequency standard was proposed and demonstrated by the groups in Germany and USA (Udem *et al* 1999, Jones *et al* 2000). Since it used to need a huge system to realize the optical frequency standard, these works became a great innovation in the field of metrology. The Nobel Prize in Physics was awarded to Dr Hall and Dr Hansch in 2005 for the invention of the optical frequency comb. At the beginning, the ultrashort-pulse Ti:Sapphire laser systems were used as the seed-comb source (Diddams *et al* 2000, Diddams *et al* 2001). However, the long-term operation for more than 24 h was difficult with the solid-state laser-based system. Then, thanks to it being maintenance free, having high long-term stability, being compact and cost effective, the fiber-laser-based optical frequency comb has gradually become the main laser source of the optical frequency comb (Tauser *et al* 2003, Washburn *et al* 2004a, Washburn *et al* 2004b, Hundermark 2004, Hong *et al* 2008). In recent times, the fiber laser optical frequency comb has been one of the hot topics in the field of laser electro-optics (Schibli *et al* 2004, Adler *et al* 2004, Newbury and Swann 2007).

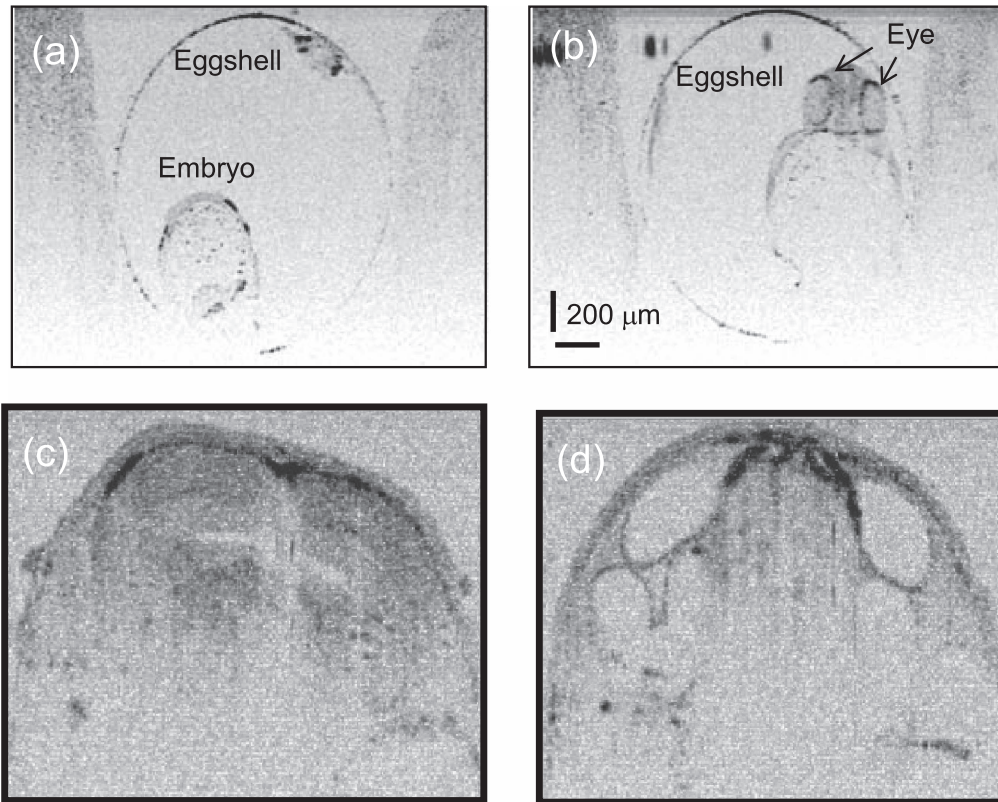


Figure 19. Cross-sectional images observed by UHR-SD-OCT with an SC at 0.8 μm wavelength: (a), (b) developing zebrafish egg, and (c), (d) head of swimming medaka (Hattori 2015).

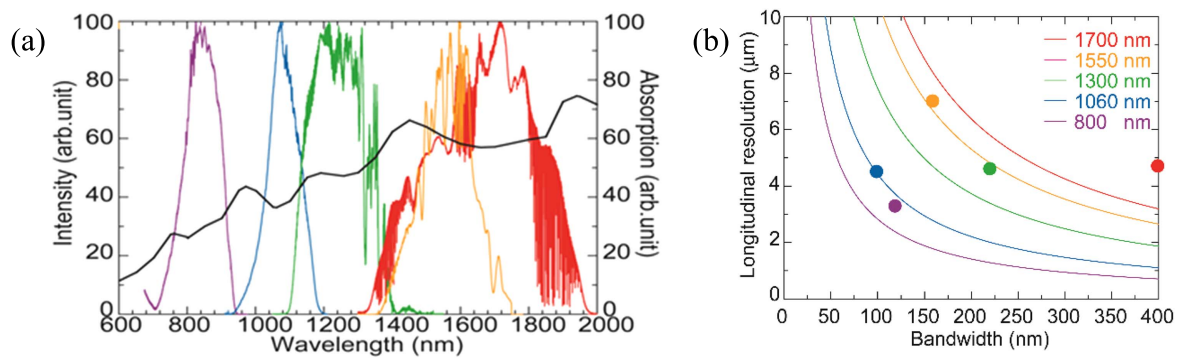


Figure 20. (a) Spectra of the generated SC in the 0.8–1.7 μm wavelength region, and (b) the relation of longitudinal resolution and bandwidth of the light source. Symbols represent experimental values (Ishida 2012).

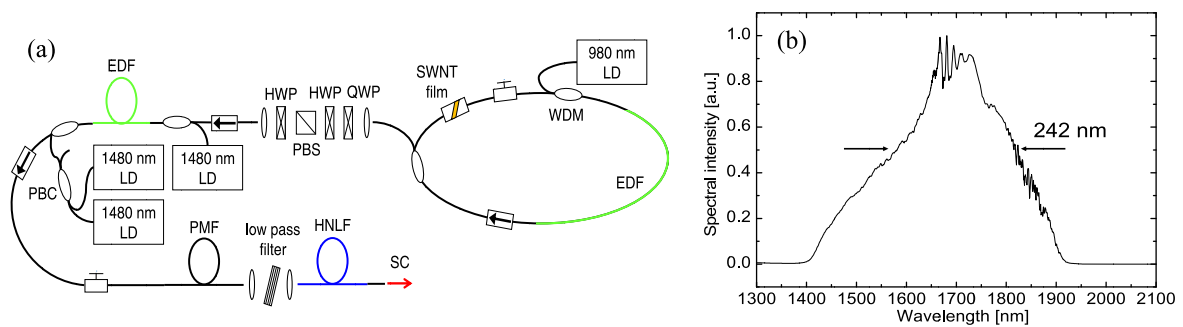


Figure 21. (a) Configuration of a 1.7 μm SC source based on SWNT fiber laser, and (b) the spectrum of the generated SC (Kawagoe 2014).

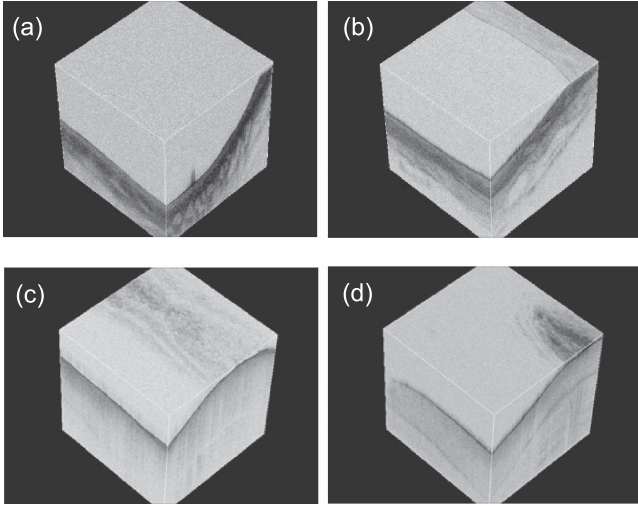


Figure 22. Observed UHR-OCT images at wavelengths of (a), (c) $0.8\ \mu\text{m}$ and (b), (d) $1.7\ \mu\text{m}$, for (a), (b) a pig trachea, and (c), (d) a human baby tooth.

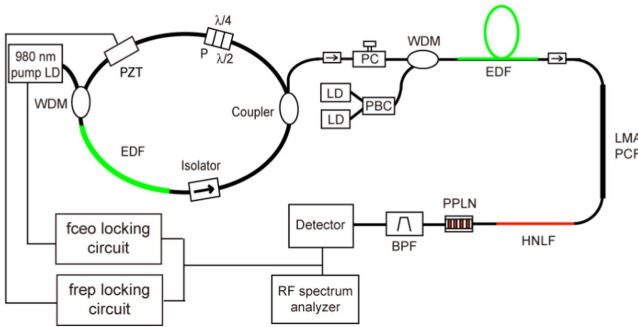


Figure 23. Setup of the optical frequency comb system based on Er-doped ultrashort-pulse fiber laser. BPF, band pass filter; HNLF, highly nonlinear fiber.

Figure 23 shows an example of an all-fiber optical frequency comb system. In the system shown in figure 23, the ultrashort pulse was amplified in a fiber amplifier, and dispersion compensation was carried out in a laser metal fusion (LMF)-PCF. Then, the generated high-power ultrashort pulse was introduced into a highly nonlinear fiber, generating a $1.0\text{--}2.0\ \mu\text{m}$ octave-spanning SC. In order to stabilize the optical frequency comb, an $f\text{--}2f$ interferometer is generally used. In the $f\text{--}2f$ interferometer, the generated octave-spanning SC is focused into a nonlinear crystal, such as PPLN, where an SHG signal of the longer wavelength component at $2.0\ \mu\text{m}$ is generated and made to overlap with the shorter wavelength component at $1.0\ \mu\text{m}$, thus generating an f_{ceo} beat signal.

Figure 24 shows the example of observed RF spectra of the f_{ceo} beat signal in a fiber-laser-based optical frequency comb system. A sharp f_{ceo} RF beat signal was clearly observed with a high signal-to-noise ratio. This f_{ceo} beat signal was then introduced into the control circuit, and the pump current of the laser diode used in the fiber laser oscillator was feedback-controlled, to stably lock the f_{ceo} . When the phase locking was achieved, the linewidth of the

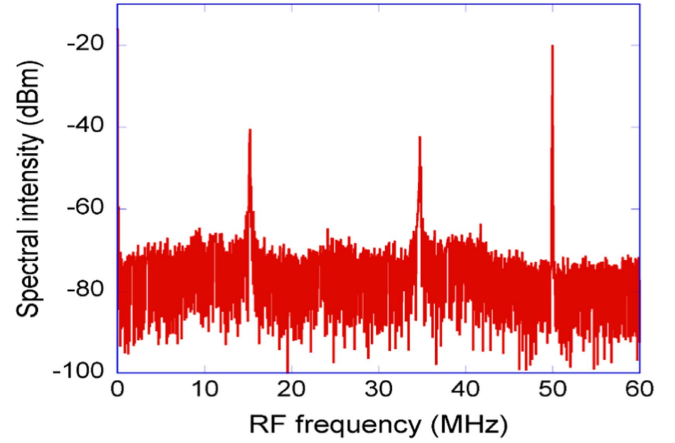


Figure 24. Observed RF spectra of the f_{ceo} beat signal with an $f\text{--}2f$ interferometer.

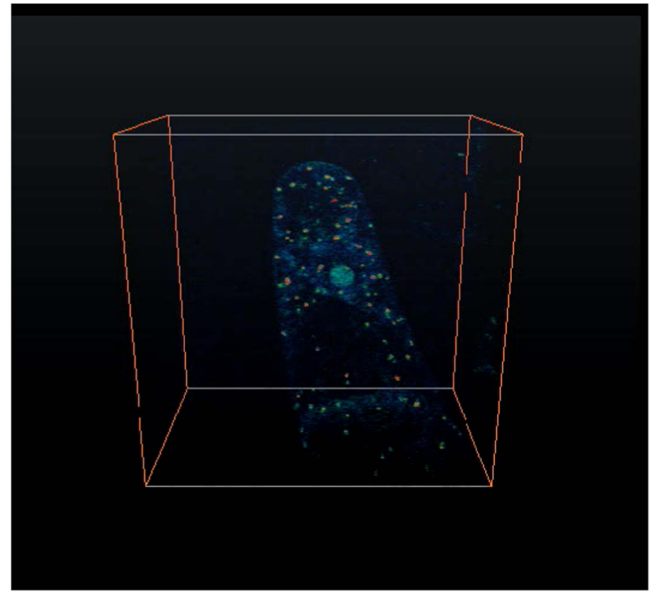


Figure 25. Observed Tabaco cell using stimulated Raman scattering microscopy with a fiber laser (Ozeki 2012).

controlled f_{ceo} beat signal was below the resolution of the RF spectrum analyzer. Fiber-laser-based optical frequency comb systems are currently used as the length standard in many countries. Recently, all-polarization-maintaining-type fiber laser frequency comb has been investigated for out of lab use (Baumann *et al* 2009, Sinclair *et al* 2014, Kuse *et al* 2015, Kuse *et al* 2016).

Fiber-laser-based optical frequency combs have also enabled breakthroughs in the field of metrology. Optical frequency combs are now used in various metrological applications, such as highly accurate length measurement, high-resolution spectroscopy, astro-combs, breath analysis, etc (Diddams 2010, Newbury 2011). Since absorption is high in the MIR region, the development of an MIR comb is one of the hot topics in the field of laser spectroscopy (Schliesser *et al* 2012).

Since a fiber laser is a compact, stable, and practical light source, it is widely used in the field of biomedical imaging.

As mentioned in the previous sections, a fiber-laser-based SC is used as a wideband light source for UHR-OCT. An ultrashort-pulse fiber laser has also been used in nonlinear microscopy. Stimulated Raman scattering microscopy, in which the spectroscopic information of molecules can be observed without fluorescent dye, has been demonstrated using synchronized ultrashort-pulse lasers operating at two different wavelengths (Freudiger *et al* 2008). Fiber-laser-based compact systems have been demonstrated by a few groups (Ozeki *et al* 2012, Freudiger *et al* 2014). Figure 25 shows the observed image of stimulated Raman scattering microscopy with a fiber laser.

Coherent anti-Stokes Raman scattering (CARS) microscopy is another nonlinear microscopic imaging technique (Zumbusch *et al* 1999, Hashimoto *et al* 2000). CARS imaging using a wideband SC source has been demonstrated by some groups (Paulsen *et al* 2003, Konorov *et al* 2004, Kee and Cicerone 2004, Kano and Hamaguchi 2005, Kano and Hamaguchi 2006). Recently, three-photon microscopic imaging was demonstrated using a high-power ultrashort soliton pulse (Wang and Xu 2011). A high-power red-shifted soliton pulse was generated with a rod-type PCF. Since the magnitude of scattering in tissue decreases as the wavelength is increased, enhancement of the penetration depth is expected. *In vivo* deep imaging of a rat brain was demonstrated using a developed high-power soliton source (Horton *et al* 2009). Label-free multiphoton imaging using an SWNT-based fiber laser system has also been reported (Kieu *et al* 2013).

7. Summary and future prospects

In this topical review paper, wideband ultrashort-pulse fiber lasers and their application to OCT and metrology are reviewed.

In section 2, Er-doped ultrashort-pulse fiber lasers, especially using SWNTs, are described. Using SWNTs, transmission-type saturable absorber devices can be developed. They are useful for constructing an all-polarization-maintaining ultrashort-pulse fiber laser, which is environmentally stable. By employing dispersion management in the cavity, several different kinds of mode-locking operations can be obtained. High-power operation was achieved in the dissipative soliton mode-locking regime.

In section 3, widely wavelength-tunable pulse sources using fiber nonlinear effects are described. Using stimulated Raman scattering and the soliton effect, we can generate ultrashort soliton pulses with rapid and wide wavelength tunability. Wideband spectral compression was demonstrated using a quasi-dispersion-increasing fiber.

In section 4, the generation of a widely broadened supercontinuum (SC) was discussed. Using a highly nonlinear fiber and ultrashort pulses, we can generate a widely broadened SC. In particular, if we use high-quality ultrashort pulses and a highly nonlinear fiber with normal-dispersion properties, we can generate a coherent, high-quality SC.

In section 5, the application of an SC to UHR-OCT is discussed. OCT is a non-destructive, micrometer-resolution,

cross-sectional imaging technique. The depth resolution is determined by the bandwidth of the light source, and UHR can be achieved using a wideband SC. Highly sensitive, UHR 2D and 3D images of biomedical samples were observed. The wavelength dependence of OCT imaging has been investigated mainly for achieving a deep penetration depth.

In section 6, the application of wideband ultrashort-pulse fiber laser sources to metrology and imaging was discussed. One of the important applications is an optical frequency comb. The use of optical frequency combs has resulted in breakthroughs in the field of metrology. Fiber lasers are mainly used as a reliable, stable, and practical pulse source for optical frequency combs. An octave-spanning SC is used for monitoring the offset frequency of an optical frequency comb. Fiber-laser-based wideband sources are also used as light sources in nonlinear laser microscopy.

Fiber laser sources are compact, stable, reliable, and promising laser sources. Wideband fiber laser sources are new, highly functional, broadband light sources that have high potential for application to imaging and metrology. Their wavelength region is being extended to both the shorter and the longer wavelength sides. In addition to advancing the field of laser applications, the development of novel technologies may be expected in the near future.

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